

Digital Arts Program - Combining Art and CS in a 50/50 Balance in Undergraduate Curriculum

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Figure 1: The Digital Arts major at UNCW combines art and computer science.

Abstract

The B.A. in Digital Arts is an interdisciplinary degree fusing studio art and computer science in equal parts. It is designed to prepare students to work in computer-art and design-related fields such as mobile-application and web design, game development, animation, visual effects, film, interactive media, graphics, simulation, human-computer interaction and other emerging opportunities at the cross-section of computer technology and the arts. This paper covers lessons learned on how best to prepare students to be successful pursuing curriculum and job opportunities in two very different, but complimentary fields.

CCS Concepts

• **Applied computing** → Education; Media arts; • **Computing methodologies** → Computer graphics; • **Human-centered computing** → Interaction design.

Keywords

Interdisciplinary curriculum development, program assessment

ACM Reference Format:

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1 Introduction

The Digital Arts (DA), B.A. degree at the University of North Carolina Wilmington is an interdisciplinary major with courses in the Departments of Computer Science and Art and Art History and was established in 2017. The curriculum is a 50/50 split between CS and Art where students study the traditional core of both fields (including programming, algorithms, and calculus for CS and drawing, graphic design, and art history for art) as well as a variety of classes that combine the two fields including 3D modeling and animation, gaming and virtual reality, digital visual effects, mobile app development, digital illustration, projection mapping, and laser cutting and 3d printing (Fig. 2).

The DA program develops students' technical, aesthetic, analytical and problem-solving skills, preparing them to work in mobile application and web design/management, UI/UX, product design, graphic design, game development, tech consulting, 3D modeling and animation, VFX, interactive media, and AI engineering. Our students also pursue graduate school programs varying from MFA's in digital media and interactive storytelling to PhD programs in CS and graphics.

As UNCW's Digital Arts major approaches its tenth year, this paper covers lessons learned in how to format an interdisciplinary program that maintains a 50-50 balance between two different fields, ensure learning outcomes from fields and the cross-section of the two are met, support students with diverse interests and backgrounds for navigating a wide variety of curriculum, and prepare students for a wide range of post-graduation opportunities.

2 Curriculum and Advising

This interdisciplinary program draws in students who may not be interested in traditional computer science or studio art degrees, but are tackling the core of these fields. Since the DA major was established, it has leveled off with about 80-100 declared majors

DA MAJOR REQUIREMENTS 71 Hours	CORE DIGITAL ARTS 10 Hours
CORE STUDIO ART 15 hours	DART 101 Introduction to Digital Arts
ART 101 Two-Dimensional Design (3)	ART 220/CSC 220/FST 220 3-D Computer Graphics Tools and Literacy
ART 111 Drawing Fundamentals (3)	CSC 368 Visual Design for Multi-Media
ART 211 Beginning Life Drawing (3)	DART 496 Digital Arts Capstone
ART 260 Introduction to Graphic Design (3)	
ART 282 Introduction to Digital Photography (3)	
CORE ART HISTORY 6 hours	MATH 8 hours
ARH 200-level Survey course (3)	MAT 161 Calculus with Analytical Geometry I (4)
ARH 300-level Digital Photography or Graphic Design (3)	MAT 162 Calculus with analytical Geometry II (4)
CORE COMPUTER SCIENCE 20 hours	ELECTIVES 12 hours
CSC 131 Introduction to Computer Science (4)	Students choose four courses from electives in the following subjects:
CSC 133 Discrete Mathematical Structures (3)	• Computer Animation
CSC 231 Introduction to Data Structures (4)	• Gaming and Virtual Reality
CSC 331 Object-Oriented Programming and Design (3)	• Digital Visual Effects
CSC 365 Client-Side Web Development (3)	• Mobile App Development
CSC 370 Computer Graphics Programming (3)	• Digital Illustration
	• Drawing
	• Graphic Design
	• Projection Mapping
	• Laser Cutting and 3D Printing
	• Game Engines

Figure 2: The DA major has students study the core of CS and of Art as well as the cross-section of the two.

any given year with an additional 20-30 students University College freshman program having indicated an interest in DA (Fig. 3). (UNCW has close to 20,000 students overall.) The DA major consistently draws in 100+ extra students to our computing classes on top of CS majors each year ($\approx 25\text{--}33\%$ increase over our ≈ 400 CS majors) and to our art classes ($\approx 100\%$ increase over ≈ 100 Studio Art majors). Over the life of this program we have developed advising materials to prepare students with very different levels of experience for these courses.

The DA program is overseen by a DA Faculty Advisory Committee (DAFAC) comprised of 3 CS faculty and 3 Studio Art faculty that meets 2-4 times per semester. The DA Program Coordinator with the support of the DAFAC has developed formal advising materials that are shared with academic advisors, students as they declare the DA major, current majors when advising for registration begins, interested perspective students/families, and on the UNCW website. These materials answer commonly asked questions and roadblocks encountered while navigating the program. The goal of these materials has been to 1) make sure all DA advisors are working with the same information and 2) to help students find the right program for their goals and efficiently move through the curriculum and address topics such as:

- Program design and goals (including why math is required for an "art" degree along with fun visual and interactive examples of math at work in DA)
- Pre-requisite and course sequences (including how to spread out the "harder" classes to avoid overwhelming semesters)
- General Education course suggestions (such as taking Anatomy to fulfill Science requirements but also help in the study of character animation)
- Equipment recommendations (what computer to buy)
- On-campus Digital Arts labs
- Declaring and changing majors (including what majors/minors complement DA for those looking for additional degrees, (i.e. Film Studies, Marketing, Creative Writing) or those wanting to switch out of DA and use completed credits)
- Degree audit software
- Course offering plans (which classes are typically offered each semester)

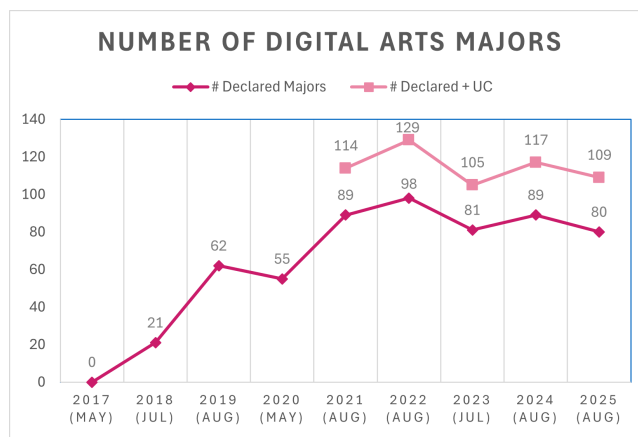


Figure 3: DA major enrollment including Freshman University College (UC) students interested in DA.

- Job/internship resources (on campus Career Center services and job titles/places of work of alumni) and
- Senior Capstone information (what types of projects are generally approved and examples of prior strong student work)

Our advising materials have aided in stabilizing major numbers with fewer students switching in and out of DA. During the 24-25 academic year, 30 new students declared the DA major and 7 switched out of DA to something else. 5 of those 7 changed majors before starting DA core classes or after taking an intro CS class meaning that through strong advising, we avoid students wasting time getting too far into a major that is a bad fit.

3 Alumni Feedback

UNCW's DA major is now old enough that we have a few cohorts of students who have completed the major, started their careers with the background outlined in Fig. 4, and have enough experience to come back through an Alumni Advisory Panel to give feedback on our curriculum design. Our panel meets once a year and alumni who live far away and are not able to attend are invited to fill out an online survey around the time of this annual meeting.

Our alumni have reported that the main strengths of the existing curriculum lie in the exposure to so many topics that are used together in industry (like programming and visual design) and the technical training that comes with learning so many different pieces of software in a short amount of time. Our alumni have reported that their programming classes helped them land jobs like web manager and graphic design and UI/UX positions over someone who did not have any CS background. They are able to speak easily with technical teams and developers and can more easily produce visual designs that will be implemented on a screen. Our alumni have also experienced having to learn new tools on the job and had the confidence to do so because they had already picked up a lot of software quickly during undergrad. Though they might not be using the exact same toolsets, they see a lot of similar programs and it simplifies onboarding. We also have had several alumni gracefully transition between more creative roles (digital content creation or



Figure 4: Skills covered in DA curriculum

VFX) and more technical roles (AI engineering). The content of the DA major has prepared them to move between diverse different career opportunities as they arise.

The primary topic our alumni say needs to be more deeply ingrained and addressed earlier in the curriculum is portfolio development. They felt they needed more coverage on how to make a more professional portfolio to compete with people with experience. They also acknowledged that undergraduate student work is often not professional enough. Alumni suggested students add case studies to their portfolios and out of class projects (such as mock websites for theoretical clients) to show willingness to go above and beyond and to build up a body of work. Alumni have reported that there are a lot of job options and current DA students should

start thinking now about what they like the best and try to focus projects in a primary area.

Feedback from advising and alumni has led us to develop a DA freshman seminar style course to share more of this information with students earlier so they can use their time in undergrad efficiently. This course includes students creating a portfolio, even if it is initially a mostly empty shell and placeholder. The idea is to just create it early and have a spot to save completed work and refine over the coming years. Our freshman seminar also includes a mentor breakfast where freshman are paired with senior capstone students and an alumni panel. Both events allow freshman to ask questions of and learn from the experiences of students who have completed the DA program already..

4 Capstone and Showcase

DA majors spend their last semester and a half at UNCW completing a capstone project. The main goal of a Capstone is to complete an original project that incorporates a balance of artistic design and computing development and demonstrates that you can integrate the different courses you have completed. Students usually work in gaming and VR, web design, app development, 3D animation, or interactive installations. Whatever the medium, capstone students are tasked with using their chosen technology to tell a story. Students generally choose something interesting and personally valued like a story inspired by their family heritage, a lifelong hobby, or plans for a dream business they hope to one day open.

The DAFAC perform advisory roles on these capstone projects by giving in-progress project feedback during the course of the semester and by attending and assessing final oral defenses at the end of each semester.

Each fall and spring semester since Fall 2021, the Department of Computer Science has held the Computing Showcase the day after classes end for the semester. Digital Arts students do a public display of their capstones during this event. The Showcase draws in visitors from current lower-level students to college faculty and administrators to high school teachers and students in the community. It is a wonderful opportunity of our current students to show off their work and to inspire the next cohort of DA majors.

5 Conclusion

The UNCW Digital Arts major has gotten off to a strong start over the last decade. The main drivers of its success have been in the interdisciplinary collaboration among faculty in developing curriculum and student opportunities and working to make sure students understand the goals of the program and how all the pieces fit together as early as possible. It remains a flexible, adaptable program that is evolving as our alumni and industry partners provide feedback and the field of Digital Arts grows and changes.

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